Ryan Diaz

Ph.D. Student, Department of Computer Science, Rice University
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Education

Rice University — Houston, TX
Doctor of Philosophy, Computer Science — Aug. 2025 – Present
- **Relevant Coursework:** Probabilistic Algorithms and Data Structures, Reinforcement Learning

University of Minnesota, Twin Cities — Minneapolis, MN
Bachelor of Science, Computer Science and Mathematics — Sep. 2021 – May 2025
- **Honors:** *summa cum laude* with high distinction (GPA 4.0/4.0)
- **Relevant Coursework:** Deep Learning for Robot Manipulation, Machine Learning, Data Analysis, Computer Vision, Natural Language Processing, Stochastic Processes, Artificial Intelligence, Linear Algebra, Probability and Statistics

Research Experience

Human-AI Teaming Lab, Rice University — Houston, TX
Graduate Researcher advised by Prof. Vaibhav Unhelkar — Aug. 2025 – Present
Topics: lifelong learning for human-robot interaction and collaboration

Robotics: Perception and Manipulation Lab, University of Minnesota — Minneapolis, MN
Undergraduate Researcher advised by Prof. Karthik Desingh — Dec. 2022 – May 2025
Topics: multisensory contact-rich robotic manipulation, imitation learning

CERL Lab, Washington University in St. Louis — St. Louis, MO
Undergraduate Researcher (NSF REU) advised by Prof. Yevgeniy Vorobeychik — May 2024 – Aug. 2024
Topics: reinforcement learning and knowledge distillation for autonomous driving

Awards and Honors

Rice Engineering Alumni (REA) Graduate Student Travel Grant Award	2026
NSF Graduate Research Fellowship Program (GRFP) Award	2026
IROS Student and Developing Countries (SDC) Travel Award	2025
UMN College of Science and Engineering Dean's List (x8)	2021 – 2025
CRA Outstanding Undergraduate Researcher Award (Honorable Mention)	2025
UMN Undergraduate Research Opportunities Program (UROP) Award (x2)	2023 – 2024
UMN Hans H. Dalaker Mathematics Scholarship Award	2024
UMN Hopper-Dean Foundation Computer Science Scholarship Award	2024
UMN Ella Thorp Mathematics Scholarship Award	2023
UMN Undergraduate Research Scholarship (URS) Award	2023

Publications

Conference Publications

[C1] **Hierarchical Reward Design from Language: Enhancing Alignment of Agent Behavior with Human Specifications**
International Conference on Autonomous Agents and Multiagent Systems (AAMAS), 2026
Zhiqin Qian, **Ryan Diaz**, Sangwon Seo, and Vaibhav Unhelkar

[C2] **AugInsert: Learning Robust Visual-Force Policies via Data Augmentation for Object Assembly Tasks**
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2025
Ryan Diaz, Adam Imdieke, Vivek Veeriah, and Karthik Desingh

[C3] **Evaluating Robustness of Visual Representations for Object Assembly Task Requiring Spatio-Geometrical Reasoning**
IEEE International Conference on Robotics and Automation (ICRA), 2024
Chahyon Ku, Carl Winge, **Ryan Diaz**, Wentao Yuan, and Karthik Desingh

Peer-Reviewed Workshop Papers

[W1] **Towards Lifelong Learning of Object-Oriented Preferences for Personalized Robotic Assistance**
Workshop on Learning from Corrections and Interventions @ Conference on Robot Learning (CoRL), 2026
Ryan Diaz and Vaibhav Unhelkar

Presentations

CoRL 2026 Learning from Corrections and Interventions Workshop, Oral/Poster — Austin, TX
Towards Lifelong Learning of Object-Oriented Preferences for Personalized Robotic Assistance — Nov. 2026

IEEE/RSJ IROS 2025, Oral/Poster — Hangzhou, CN
Learning Robust Visual-Force Policies via Data Augmentation for Object Assembly Tasks — Oct. 2025

WashU STEM Poster Palooza, Poster — St. Louis, MO
Vision-Based Algorithms for Obstacle Detection and Avoidance in Autonomous Vehicles — Aug. 2024

UMN Undergraduate Research Symposium, Video — Minneapolis, MN
Augmenting a Dual-Arm Contact-Rich Robotic Manipulation Task with Force-Torque Data — Dec. 2023

UMN Summer Undergraduate Research Expo, Poster — Minneapolis, MN
Imitation Learning for Spatio-Geometry Driven Assembly Task with Dual-Arm Manipulator — Aug. 2023

UMN Undergraduate Research Symposium, Poster — Minneapolis, MN
Large-Scale Object Generation for Learning Robotic Manipulation Tasks — Apr. 2023

Teaching Experience

COMP 459/559: Machine Learning with Graphs — Spring 2026
Graduate Teaching Assistant, Rice University

CSCI 4521: Applied Machine Learning — Spring 2025
Undergraduate Teaching Assistant, University of Minnesota, Twin Cities

MATH 5652: Introduction to Stochastic Processes — Spring 2025
Undergraduate Paper Grader, University of Minnesota, Twin Cities

CSCI 2033: Elementary Computational Linear Algebra — Spring 2023, Fall 2023, Fall 2024
Undergraduate Teaching Assistant, University of Minnesota, Twin Cities

CSCI 4511W: Introduction to Artificial Intelligence — Spring 2024
Undergraduate Teaching Assistant, University of Minnesota, Twin Cities

CSCI 1933: Introduction to Algorithms and Data Structures — Summer 2023
Undergraduate Teaching Assistant, University of Minnesota, Twin Cities

UMN Taylor Tutoring Center — Fall 2022
Undergraduate Peer Tutor, University of Minnesota, Twin Cities

Academic Service

Peer Reviewing

Conference Reviewer: ICRA (2025), IROS (2025, 2026)
Workshop Reviewer: CoRL Workshop on Learning from Corrections and Interventions (2026)
Undergraduate Peer Reviewer: Minnesota Undergraduate Research and Academic Journal (2024, 2025)

Mentoring

Terence Li (Undergraduate @ Rice University) Oct. 2026 – Present
Project: Investigating Human Communication Strategies in Long-Term HRI

Gael Cisneros (High School Student) Jun. 2026 – Aug. 2026
Project: Facilitating More Natural HRI with a Robot Face

Shubham Panchal (High School Student) Jan. 2026 – May 2026
Project: Few-Shot Manipulation via Task-Aware Model-Agnostic Meta-Learning

Technical Skills

Languages: Python, Java, C, C++, JavaScript, HTML, MATLAB, Bash, LaTeX
Libraries: PyTorch, TensorFlow, OpenCV, NumPy, ROS(2), Transformers (HuggingFace), MuJoCo
Software: Git, Linux, Blender